



BICOL UNIVERSITY
COLLEGE OF SCIENCE
Legazpi City, Albay
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Individual Journal

Name: Edgar Neil M. Bayoca

Group Name: Project Bloom

Role in the Group: (e.g., *Backend Developer, Client Developer, Tester, etc.*)

Topic / Activity: Laboratory 4: REST API

Date Submitted: April 16, 2026

Overview of My Role

I served as the Code Reviewer, Backend Assistant, and QA Tester for Project Bloom. My primary focus was ensuring the integrity and quality of the codebase by reviewing peer contributions, checking the GitHub repository structure, and monitoring commit history for discrepancies. Additionally, I assisted in backend development with a specific focus on security and data validation to ensure the REST API remained robust against invalid inputs and edge cases.

Specific Tasks Performed

Conducted thorough code reviews to ensure clarity, readability, and adherence to project standards.

Audited the GitHub repository structure and monitored commit history to maintain project organization and transparency.

Performed Quality Assurance (QA) testing on API endpoints to identify potential system errors and edge cases.

Validated backend input logic for student registrations and grade entries.

Tested the system's limitations to prevent data corruption from invalid user inputs.

Code Contributions

I focused on the validation layer of the backend PHP code. I implemented the logic for `isValidGrade` to restrict grade entries to the official scale and created validation rules for user registration, such as minimum username lengths and password constraints. These contributions ensure that the data processed by the API is accurate and that the system returns appropriate HTTP 400 (Bad Request) or 409 (Conflict) errors when invalid data is submitted, rather than breaking the application.

Challenges Encountered

A major challenge was identifying and mitigating system-breaking inputs. During the testing phase, I discovered that the system could crash or the database could be corrupted if users submitted non-numerical data (like text instead of a grade) or duplicate records. Managing these edge cases while maintaining a smooth user experience across different clients (the Web and Python applications) was difficult, as each client required consistent error responses from the backend.

Problem-Solving and Contributions

To resolve these issues, I collaborated with the backend team to integrate strict server-side validation checks directly into the API logic. By testing various "worst-case" inputs, I helped define the system's limitations and refined the error-handling logic to prevent crashes. This collaborative effort led to a more stable API that provides clear feedback to both the web and console applications, preventing accidental or malicious data corruption.